

Technical Presentation

Intro

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Lightsabers

In Star Wars

- mighty weapon that all jedi and siths possess
- one of the most recognisable weapon of pop culture
- saber that emits light
- can cut through most materials
- retractable
- can block other sabers
- can be of any color
- very hot
- works with kyber crystals

Solid Saber

Let's start simple. Let's make a solid light saber.

We need a solid material that can withstand enormous heat, for example titanium. Titanium is easy to heat but melts quite easily at high temperatures. The solution is put the titanium around a core of some other material like tungsten.

These materials are very conductive, so we just need some electricity to heat them up.

When heated up, these materials naturally glow and emit light. Sadly, the color can't really be changed to anything else than red or orange.

Still, the main problem is that the blade can't be retracted. It would just be a really hot knife.

Laser Saber

Why not use lasers as the name implies?

Lasers are hot and can cut through most materials -given enough energy. They emit light and can be of any wanted color. Plus, once the saber is turned off, we're just left with hilt as in the movies.

However, a laser is just a light beam. It does not stop at any fixed length. That means our real life Jedi would have infinite reach. In addition, two such light sabers would not block each other. They would just pass through each other.

Basically such a technology would make the light sabers act like very very powerfull flashlights. Nothing more.

Plasma

So there is a third and less obvious option left: plasma.

Plasma is a state of matter in which the atoms are so hot that they lose their electrons. This fluid is electrically charged and can be manipulated with magnetic fields. Actually, we all know plasma: lightning is a form of plasma.

So that's it. We just need to create a plasma beam and use magnetic fields to keep it in the shape of a blade. The problem is that the magnets would need to be close to the blade to keep it in shape. That would defeat the purpose of a light saber.

Plasma Flame Saber

But plasma can be created in another way: by burning gas. Basically, we’re building a powerfull flame thrower.

This model would have most of the properties of a light saber. It would be very hot, cut through most materials, and be retractable.

Changing the gas would allow us to change the color of the flame to will.

However the blade would not be solid and a real life light saber battle would be impossible.

This is as far as our technology can go. We can create a light saber that looks like the one in the movies, but it would not be able to block other sabers.

The youtube channel Hacksmith Industries has created multiple prototypes of light sabers. Their last one is based of the plasma flame model.

Conclusion

characteristics technologies	solid	laser	plasma	plasma flame
~1m blade	OK	KO	KO	OK
various colors	KO	OK	OK	OK
hot	OK	OK	OK	OK
cuts all	KO	OK	OK	OK
block other sabers	OK	KO	KO	KO
emits light	OK	OK	OK	OK
retractable	KO	OK	OK	OK

Showcase 1

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Showcase 2

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Translator Droids

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Transhumanism: prosthesis

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Conclusion

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